# Homework 2: Probability and Uncertainty

### POLS 602

Due: October 30, 2026

## Instructions

Submit both your code and answers by pushing them to your github repo. They may be in separate files, or in a single file like a .qmd, .rmd, or .ipynb notebook. Comment your code and made sure it executes without errors.

## Problem 1: Simulation

Create a simulated dataset in which the outcome is a linear function of a treatment variable and a confounding variable. Fit a linear model for the true data generating process and print the summary table. Then do the following.

**(a)** Using the true model, demonstrate that the coefficient for your treatment variable follows the central limit theorem. That is, demonstrate that the coefficient's sampling distribution is approximately normal.

**(b)** Compute the bootstrapped standard error for the coefficient of the treatment variable.

**(c)** Fit a model that omits the confounding variable. Repeat (a) for this new model and plot the sampling distribution of the treatment variable's coefficient. How do your results differ? What does this imply about statistical tests based on a coefficient's sampling distribution?

## Problem 2: Data Analysis

Use any dataset you like. It can be from the course materials, simulated, or drawn from another source.

**(a)** Conduct a hypothesis test for a difference in means. You decide what the hypotheses are, whether to use a t-test or a z-test, and the level of significance. Explain your decisions, and interpret your results both substantively and statistically.

**(b)** Using the same data, fit a linear model. Interpret the coefficient, standard error, t-value, and p-value.

**(c)** Conduct a permutation test for your hypothesis. Plot the null distribution of the test statistic and mark your observed test statistic on the plot. Discuss your results, including how the permutation p-value compares with the p-values from (a) and (b).